STEP 1: Create & Use Database

```
use UniversityDB
```

Check:

```
show dbs
```

STEP 2: Create Collections

```
db.createCollection("students")
db.createCollection("courses")
db.createCollection("teachers")
```

Check:

```
show collections
```

STEP 3: Insert Dummy Data

Students Collection

```
db.students.insertMany([
  {
    name: "Ali Khan",
    age: 20,
    department: "Computer Science",
    semester: 4,
    GPA: 3.5,
    fullTime: true,
    courses: ["DBMS", "OOP", "DSA"],
    address: { city: "Peshawar", zip: 25000 },
    admissionDate: new Date("2023-09-01")
  },
  {
    name: "Sara Ahmed",
    age: 22,
    department: "Software Engineering",
    semester: 6,
    GPA: 3.9,
    fullTime: true,
    courses: ["AI", "ML", "DBMS"],
    address: { city: "Mardan", zip: 23200 },
    admissionDate: new Date("2022-09-01")
  },
  {
    name: "Usman Ali",
    age: 19,
    department: "Computer Science",
    semester: 2,
    GPA: 2.8,
    fullTime: false,
    courses: ["Programming", "Math"],
```

```
    address: { city: "Dir", zip: 18000 },
    admissionDate: new Date("2024-02-01")
  }
])
```

Courses Collection

```
db.courses.insertMany([
  { courseCode: "CS101", title: "Programming", creditHours: 3 },
  { courseCode: "CS202", title: "DBMS", creditHours: 4 },
  { courseCode: "CS303", title: "AI", creditHours: 3 }
])
```

Teachers Collection

```
db.teachers.insertMany([
  { name: "Dr. Ahmad", department: "CS", experience: 10 },
  { name: "Ms. Ayesha", department: "SE", experience: 6 }
])
```

STEP 4: READ Operations (find / findOne)

```
db.students.find()
db.students.findOne({ name: "Ali Khan" })
```

Projection

```
db.students.find(
  {},
  { name: 1, GPA: 1, _id: 0 }
)
```

STEP 5: QUERY OPERATORS

Comparison

```
db.students.find({ GPA: { $gte: 3.0 } })
db.students.find({ age: { $lt: 21 } })
```

Logical

```
db.students.find({
  $and: [
    { department: "Computer Science" },
    { GPA: { $gte: 3.0 } }
  ]
})
db.students.find({
  $or: [
    { fullTime: false },
```

```
    { GPA: { $lt: 3.0 } }
  ]
})
```

STEP 6: ARRAY OPERATIONS

```
db.students.find({ courses: "DBMS" })
db.students.find({
  courses: { $in: ["AI", "ML"] }
})
db.students.find({
  courses: { $all: ["DBMS", "OOP"] }
})
```

STEP 7: SORT, LIMIT, SKIP

```
db.students.find().sort({ GPA: -1 })
db.students.find().limit(2)
db.students.find().skip(1).limit(1)
```

STEP 8: UPDATE Operations

$set

```
db.students.updateOne(
  { name: "Usman Ali" },
  { $set: { fullTime: true } }
)
```

$inc

```
db.students.updateOne(
  { name: "Ali Khan" },
  { $inc: { semester: 1 } }
)
```

$push

```
db.students.updateOne(
  { name: "Sara Ahmed" },
  { $push: { courses: "Cloud Computing" } }
)
```

$unset

```
db.students.updateOne(
  { name: "Usman Ali" },
  { $unset: { address: "" } }
)
```

STEP 9: DELETE Operations

```
db.students.deleteOne({ name: "Usman Ali" })
db.students.deleteMany({ GPA: { $lt: 3.0 } })
```